

CAPM Regression and Volatility Dynamics

Question

How do firm-level betas and volatilities (both systematic and idiosyncratic) vary across industries and over time, and how are they related to stock returns?

Data

The data used for this project was monthly CRSP stock return data from 1996 to 2023. The dataset consisted of firm identifier, SIC industry codes, and value-weighted market returns as a proxy. I randomly selected ten firms per industry each year to reduce computational intensity.

Approach

My project estimated firm-level CAPM by performing time-series regressions of individual stock excess returns on markets returns. A stock's market beta can be estimated using the following CAPM or one-factor market model regression:

$$r_{i,t} = \alpha_i + \beta_i \cdot MKT_t + \epsilon_{i,t}$$

where $r_{i,t}$ is the excess return of stock i during period t , MKT_t is the excess return of the market factor or portfolio during period t , and $\epsilon_{i,t}$ is the regression residual. This estimation involved performing a linear regression, where the dependent variable was the excess return of stock i , and the independent variable was the excess return of the market. Betas were approximated using rolling windows of 12, 24, and 36 months. Descriptive stats such as mean, standard deviation, skewness, kurtosis, and percentiles were computed for each industry-year combination.

Given the market model estimation, using the regression residuals, I then decomposed total volatility into two components, systematic and idiosyncratic volatility, as follows:

$$(\text{var}(R_{i,t}))^{1/2} = \beta_{i,t}(\text{var}(R_{m,t}))^{1/2} + (\text{var}(\epsilon_{i,t}))^{1/2}.$$

That is, the total volatility of stock i , $TVOL_{i,t} = (\text{var}(R_{i,t}))^{1/2}$, is the sum of the systematic volatility $SVOL_{i,t} = \beta_{i,t}(\text{var}(R_{m,t}))^{1/2}$, and the idiosyncratic volatility $IVOL_{i,t} = (\text{var}(\epsilon_{i,t}))^{1/2}$.

Lastly, I sorted stocks into quintile portfolios based on beta and idiosyncratic volatility to examine their relationship with average excess returns, both equally and value weighted.

Findings

The Services sector consistently exhibited the highest 12-month average beta among the ten industries analyzed, indicating that it was more sensitive to market movements and, on average, more volatile relative to the overall market. The “Other” industry group showed the highest standard deviation, suggesting greater dispersion and firm-level variability within that category. Over a 24-month window, the “Other” category displayed the highest average beta, implying that its firms were more exposed to systematic market risk over a two-year horizon. Meanwhile, the Construction sector showed the highest standard deviation, indicating more erratic or cyclical performance relative to the broader market. Across the 36-month estimation, the “Other” industries again maintained the highest average beta, confirming its persistent market sensitivity. The Construction sector once more exhibited the highest standard deviation of beta values, highlighting its ongoing volatility.

The three measures of volatility—total, systematic, and idiosyncratic—followed broadly similar time patterns throughout the sample period. As expected, systematic volatility reflects market-wide risk factors that affect all firms, while idiosyncratic volatility captures firm-specific risk. The time-series graphs show that idiosyncratic volatility consistently exceeded systematic volatility, highlighting that firm-level uncertainty typically outweighs aggregate market risk in magnitude. Total volatility represents the combination of both components. During market downturns or crises, systematic volatility tended to rise sharply, narrowing the gap with idiosyncratic volatility—demonstrating how firm-level and market-level risks converge under stressed economic conditions.

Computational Environment

The data analysis and computations were computed using Python in a Python notebook environment. The main Python libraries used were Pandas, NumPy, Statsmodels, Scipy, and Matplotlib. Pandas and NumPy were mainly used for data handling and manipulation. Statsmodels was used for OLS regression. Scipy was used to examine the skewness and kurtosis of the betas. Matplotlib was used to plot and display time series graphs of the computed data.